DOTE 2030 Midterm Cheatsheet · Lectures 2–5

1 · Process mapping (L2)

- **Flow unit**: item flowing through. **Activity** (□): takes time, uses resources. **Buffer** (△): waiting. **Decision point** (◇): routes flow.
- Only **resources** have capacity & can be bottlenecks, never activities. Count **every** resource: workers, machines, tools (the handsaw trap: *you* 2 h + 1 h = 3 h is the bottleneck).
- Map the process as it *actually* works; validate with the people involved.

2 · Flow time

Theoretical flow time = *shortest* time for one unit (perfect world, no waiting). Actual flow time can be longer.

Single path: sum of activity times.
Simultaneous activities: longest path START→END.

- A pooled activity (W1 or W2 does A) counts **once**.
- Burger ‖ fries: 1+5+1+1+1 = 9 min, not 13. HW1 Q1: paths 18, 20, 20, **22**.
- Pooling or speeding a non-bottleneck task changes flow time, not capacity. A faster bottleneck task changes both.

3 · Unit load · capacity · bottleneck

Unit load (resource) = Σ times of all activities it does
Capacity = (# in pool) × 60 ÷ unit load (min) per hr
Process capacity = min over resources/pools = bottleneck

- Unit load = 1 / capacity. Bottleneck = highest unit load per pool member. Ties are possible.
- An activity needing 2 resources (Crane + Emp 1) adds its time to **both** unit loads.
- **Never** compute capacity as 1 / flow time (McDonald's: 36 s ≠ 100/hr; real capacity 360/hr).

McDonald's: 8, 10, 8, 6, 2, 2 s → 450, 360, 450, 600, 1800, 1800 /hr
Flow time 36 s. 2nd cashier: still 360. 2nd toaster: **450**, new bottlenecks cashier + W2. W3 does 6+2+2 = 10 s → still 360, frees W4, W5.

Improving capacity

- (1) **Pool** the bottleneck (add copies). (2) **Reduce its unit load**: speed it up or reassign tasks.
- Fixing one bottleneck can create another. Reduce only until it ties the **next** bottleneck; watch pools (Review SA2: worker 25 → 15 min, = **10 min**).
- Equipment caps capacity whatever the staffing (pizza oven: 1+10+1 = 12 s → max 5/min). Balancing work: 4.29 → 5/min, same 22 s of total work.

4 · Throughput & utilization

Throughput = min{input rate, capacity rate}
Utilization = throughput ÷ resource capacity (≤ 100%)
Implied util. = input ÷ resource capacity (can be > 100%)
Cycle time = 1 ÷ throughput

- All utilizations < 100%: **demand-limited**. Some = 100%: that resource is the bottleneck, and input ≥ capacity.
- Implied util. > 100% → inventory builds. Utilization is **never** > 100%.
- Throughput = capacity does *not* imply build-up (input might equal capacity exactly).
- Bottleneck has the highest utilization.

Profit decisions

Profit/hr = throughput × margin − hourly costs. Only adding capacity at the bottleneck helps; the new bottleneck caps the gain. Rev. SA1(e): rent 2nd B → cap 4/hr: $40 − $14 = $26 < $30, so don't rent. SA11(c): hire W3 → 8.57/hr: $85.7 − $60 = $25.7 > $20, so hire.

5 · Worked patterns

Case	Unit loads → capacity	Answer
In-class ex.	Pool A 2×6 = 12; W3 5+2+3 = 10 min → 6; W4 5 → 12	BN W3, 6/hr; flow time 25; util @3/hr 25/50/25%
HW1 Q4	Saw 30, E1 40+10 = 50, Crane 10, E2 60	BN E2 (1/hr); pool E2+E3 → E1 (1.2/hr); target 2/hr: all ≤ 30 min → $300 + $200 = **$500**
HW1 Q5	R1 6+2 = 8 → 7.5; R2 2+4 = 6 → 10; R3 4 → 15	Util R2 @ input 2 = 20%
Bank SA11	W1 4+2 = 6 → 10; W2 3+7 = 10 → 6; PC 2 → 30	6/hr; implied util W1 @12 = 120%

6 · Inventory build-up (L3)

- Inventory in process = buffers + activities (pizza 3 + 1 + 1 = 5). Never < 0.
- End-of-interval queue: **$Q = \max\{0, Q_{prev} + input - capacity\}$**.
- Slope (build-up rate) = input − capacity while a queue exists. Output = capacity while queued, else = input.
- When the queue drains, **find when it hits 0** = Q ÷ (capacity − input).

Discrete avg = Σ snapshots ÷ # intervals
Continuous avg = area under curve ÷ total time

- Split the area into triangles (½·b·h) and rectangles. A trapezoid = rectangle + triangle.
- Same data, different methods: weeks 20 → 50: discrete (20+50)/2 = 35; continuous (10+20+15)/2 = 22.5.
- Enough *total* capacity ≠ no queue (airport: 195 ≤ 240 yet queue peaks at 13; avg 51/16 = 3.19).
- **Stable**: long-run input ≤ capacity. Short-run input can exceed it.

Example	Area	Avg
50/hr, 200/hr (12–2), cap 100, 10a–6p	½·2·200 + ½·4·200 = 600	600/8 = **75**; R = 700/8 = 87.5
HW1 Q3: +20/hr 2 h, flat 2 h, −20/hr 1 h	40 + 80 + (20+10) = 150	150/5 = **30**
Start 65, +4/hr, 5.5 h (end 87)	357.5 + 60.5 = 418	**76**
POS: 15, 32, 39, 10 vs cap 30	queues 0, 2, 11, 0	13/4 = **3.25**

Review SA8 (hospital): after 12 pm, slope = 90 − 150 = −60/hr, so 50 clears at **12:50**. Area 25 + 20.83 = 45.83 → avg **9.17** (review key says 10). Show the 12:50 step.

7 · Little's Law (L3)

I = R × T (avg inventory = throughput × avg flow time)

- Requires stability. I, R, T must cover the **same** part of the process, in the **same** time units.
- Rocks: R = 12/hr, T = 10 min = 1/6 hr → I = 2. I = 4 + 1 = 5, T = 20 min → R = 0.25/min.
- BBA: 850 × 4 = 3400; 3500/4 = 875; 3600/800 = 4.5 yr. Queue: $T_q = I_q/R$ = 75/87.5 h = 51.4 min.
- Heathrow: count in/out → I, R → T = I/R (no need to match individuals). Uber: L − O = Y(η + T).

8 · Shouldice (L3 case)

- Weekly capacity: beds 90 × 7/3 = 210; surgeons 12 × 4 × 5 = 240; **ORs 5 × 8 × 5 = 200 (bottleneck)**. Daily: ≤ 90 beds, ≤ 48 surgeons, **≤ 40 ORs**.
- Now: R = 150/wk; I = 150 × 3/7 ≈ 64. Best Mon–Fri schedule: 40, 40, 10, 40, 40 = **170** ($x_1 + (x_2+x_3+x_4) + x_5 \le 40 + 90 + 40$). 180 needs Saturday surgery, +10 beds, or +1 OR.

9 · Scheduling: SPT & WSPT (L4)

$C_i = W_i + p_i \Rightarrow \Sigma C = \Sigma W + \Sigma p$ (Σp fixed)

- **SPT**: ascending p_i → minimises ΣC and ΣW. Why: the 1st job's time appears in all n completion times.
- **WSPT**: ascending $\mathbf{p_i/w_i}$ → minimises $\Sigma w_i C_i$ and $\Sigma w_i W_i$.
- Rank by ratio, but compute C_i by adding the **actual** p_i. $\Sigma w_i W_i = \Sigma w_i C_i - \Sigma w_i p_i$.

Example	Order	Result
p = 10, 15, 7	3→1→2	ΣC 56, ΣW 24 (vs 67/35)
p = 6, 5, 8, 15	2→1→3→4	C 5, 11, 19, 34; ΣC 69; ΣW 35
p 6, 5, 8, 15; w 4, 1, 2, 1	1→3→2→4	ratios 1.5, 5, 4, 15
10-patient ex.	2,7,9,5,1,4,8,6,3,10	ΣwC 1038; ΣwW 748

10 · Variability

Same averages ≠ same queue. ATM (0.5/min, 1 min service): regular 0 wait; clustered arrivals 0.5 min; variable service 0.125 min. Flow time = waiting + service.

11 · Probability (L4)

$E[X] = \Sigma\, x_i \cdot P(x_i) \cdot \text{Var} = E[(X - E[X])^2]$
$\sigma = \sqrt{\text{Var}} \cdot \text{CV} = \sigma / E[X]$ (no units)

1, 3, 5 items with prob .25/.5/.25 → E = 3, Var = 2, σ = 1.41, CV = 47.1%.

Model	Use	Mean	SD	CV
Poisson(m)	count in a window	m	√m	1/√m
Exponential(r)	duration / gap	1/r	1/r	**1**
Deterministic	constant	c	0	**0**

- Arrivals at λ/hr: A ~ Exp(λ), E[A] = 1/λ. In 10 min, N ~ Poisson(λ/6). λ = 12: Poisson(2), CV .707.
- Poisson counts ⇔ exponential gaps. 15/hr → SD of gap = 1/15 h = **4 min**.
- Mean = SD does **not** prove the distribution is exponential.

12 · Single-server queue (L4)

λ arrival rate; 1/λ = E[A]	μ service rate; 1/μ = E[S]
$\rho = \lambda/\mu$; stable iff ρ < 1; throughput = λ	

PK: $l_q \approx \rho^2/(1-\rho) \times (C_A^2 + C_S^2)/2$
$T_q = l_q/\lambda \cdot T_s = 1/\mu \cdot l_s = \rho$
$l = l_q + l_s \cdot T = T_q + T_s = l/\lambda$

- C_A = SD(A)/E[A]; C_S = SD(S)/E[S], using the **times**, in the **same units**.
- l_q excludes the customer in service.
- Alt. form: $l_q \approx \lambda^2/[\mu(\mu-\lambda)] \times (\lambda^2\sigma_A^2 + \mu^2\sigma_S^2)/2$.

Model	C_A	C_S	l_q
G/G/1	C_A	C_S	$\rho^2/(1-\rho)\cdot(C_A^2+C_S^2)/2$
M/M/1 (exact)	1	1	$\rho^2/(1-\rho) = \lambda^2/[\mu(\mu-\lambda)]$
M/D/1	1	0	$\rho^2/(1-\rho)\cdot½ = \lambda^2/[2\mu(\mu-\lambda)]$
D/D/1	0	0	0

Notation: arrivals / service / # servers. G = general, M = exponential (CV 1), D = deterministic (CV 0).

13 · Queue worked answers

Case	ρ	l_q	T_q
λ4 μ6 M/M/1	2/3	4/3	1/3 h = 20 min
λ4 μ6 C_A 1/3, C_S .3	2/3	0.134	.034 h ≈ 2 min
λ4 μ6 M/D/1	2/3	2/3	1/6 h = 10 min
λ10 μ20 M/M/1	1/2	1/2	1/20 h = 3 min
λ10 μ20 M/D/1	1/2	1/4	1/40 h = 1.5 min
Drive-thru λ20 μ30	2/3	4/3; l = 2	4 min; T = 6 min (M/D/1: T = 4)
D/G/1 homework	5/12	25/672	≈ .0037 days

14 · Utilization vs variability · OM triangle

- M/M/1 l_q at ρ = 50/80/90/95%: **0.5 / 3.2 / 8.1 / 18.05**. 80→90% more than doubles it.
- At ρ = 80%: exponential service 3.2 vs constant service 1.6.
- No variability: max utilization is good. With variability: it's very bad (queues explode as ρ→1).
- λ = 8: 6 min exp → 24 min wait; 5 min exp → 10 min; 6 min constant → 12 min. Target l_q 1.6: M/M/1 needs μ = 11.48, M/D/1 needs 10.
- **OM triangle**: **Capacity** (capacity cost) · **Inventory**/buffer (waiting cost) · **Information** to cut variability, e.g. scheduling (improvement cost).

15 · Multi-server queue (L5)

$\rho = \lambda/(c\mu)$, stable iff $\lambda < c\mu$
$l_q \approx \rho^{\sqrt{2(c+1)}}/(1-\rho) \times (C_A^2 + C_S^2)/2$
$l_s = c\rho = \lambda/\mu \cdot T_q = l_q/\lambda$

c	1	2	3	4	5
√(2(c+1))	2	2.449	2.828	3.162	3.464

- Pool util. = each server's util. = λ/(cμ). M/M/c: $l_q \approx \rho^{\sqrt{2(c+1)}}/(1-\rho)$.
- **Pooling wins**: one G/G/c queue < c separate G/G/1 queues ($c\cdot\rho^2/(1-\rho)$…).
- Card office λ90, μ20, c5: ρ .9 → l_q 6.94, T_q 4.63 min.
- Tellers: separate (5/10) + (15/20) = ½ + 9/4 = 2.75 vs M/M/2 (λ20, μ15) ρ 2/3 → **1.11**.
- Airport: separate lines 3, 6, 30 min → weighted (28·30 + 20·6 + 10·3)/58 = 17.07 min. Merged M/M/3, ρ 58/90 → l_q .812, T = .84 + 2 = **2.84 min**.
- **Independent queues**: add the l_q; avg T_q is weighted by λ share (17/105 h). **Series**: add each T_q (helpline .003 + .462 = .465 h).

16 · Matching (L5, conceptual)

Platform decides **when** to match (now vs wait) and **whom**. Engagement comes from **suspense** and **surprise** → match similar skill. 1000/1400/1010/1390: match at once → gaps 400/380 (short wait); wait for 4 → gaps 10/10 (long wait, risk of quitting). Drawbacks: long queues for top players; win rate stuck near 50%.

17 · True/False traps (review)

F	Theoretical flow time = every unit's time
T	Single bottleneck; shorten its task → capacity ↑
F	Shorten *any* task → capacity ↑
T	Shorten R's task → R's capacity ↑
T	Implied util. > 100% → inventory builds
F	Throughput = capacity → must build inventory
F	Input > capacity → utilization > 100%
F	Spare capacity 6–7 pm → queue at 7 pm is shorter
F	Activities have capacity rates
F	l 100, T 5 h → R 500 (it is 20)
F	Utilization ↑ → l_q ↓
T	D/D/1 → l_q = 0
F	Mean = SD → must be exponential
F	Arrival variance ↑ → more people *in service*
F	Double λ and μ (same SDs) → l_q ×2 (it is **×4**)

18 · Exam checklist

- Convert everything to one time unit **before** computing ρ or CV.
- Check stability (ρ < 1) and say which model you're using (M/M/1, M/D/1, G/G/c).
- State the formula, substitute, then give the units. Round to 2 dp.
- Asked for "in system / including service"? Add l_s or T_s.